

2-Day Task: User Notification Preference & Delivery API

Objective

Build a backend service that manages user notification preferences and simulates notification delivery.

Functional Requirements

1. User Management

- Create a user (name, email, phone)
- Get user by ID
- List all users

2. Notification Preference Management

- Enable / disable notification channels per user (EMAIL, SMS, PUSH)
- Update preferences for a user
- Fetch preferences for a user

3. Notification Trigger

- Send a notification to a user with message content and notification type
- Validate user existence and enabled channel
- Store notification history with status (SENT, FAILED)

4. Notification History

- Fetch all notifications for a user
- Fetch notifications by status
- Fetch notifications by type

Business Rules

- Disabled channels must not receive notifications
- Each notification attempt must be recorded
- Default preference: all channels disabled

Technical Constraints

- Java + Spring Boot
- REST APIs only
- JPA/Hibernate
- H2 Database
- Layered Architecture

Deliverables

- Source code repository
- Sample API requests/responses
- README with run instructions

Timeline

2 Working Days